

ICT171 Documentation

Jamie Skelin 35202149

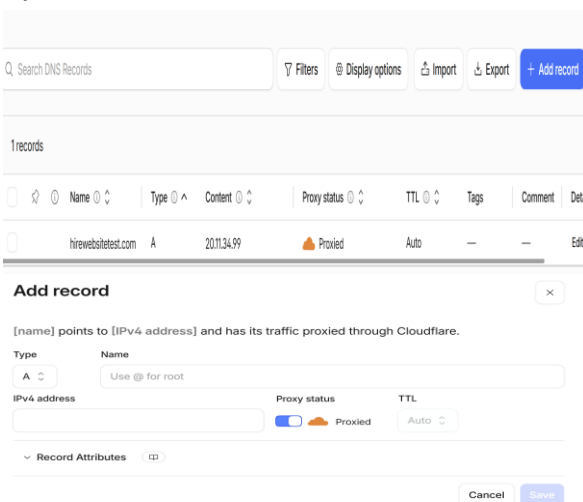
Setting up a website:

1. Go to <https://portal.azure.com/#home>
2. Sign up or login (as a student if possible for \$140 of free credits)
3. Create a VM to suite your needs (in this case a standard B2ats running Ubuntu)
4. Save the SSH key onto your local device
5. Remotely SSH into your new VM and install Nginx
 - a. Go into powershell
 - b. `ssh -I /path/to/privatekey.pem jUser@20.11.34.99`
 - c. `sudo apt update`
 - d. `sudo apt install nginx`
6. Your VM can now act as a server

Setting up DNS

1. Go to a web provider (eg <https://www.cloudflare.com/products/registrar/>)
2. Find a domain name you like and is available and purchase it
3. Inside of your providers settings create an A record pointing towards your website public IP address

a.



The screenshot shows the Cloudflare DNS management interface. At the top, there's a search bar and buttons for 'Filters', 'Display options', 'Import', 'Export', and '+ Add record'. Below this, a table lists existing DNS records. One record is visible for 'hirewebsites.com' with type 'A' and content '201134.99'. Below the table, the 'Add record' form is open. It shows the 'Type' as 'A', the 'Name' as 'Use @ for root', and the 'IPv4 address' field. The 'Proxy status' is set to 'Proxied' (indicated by an orange cloud icon) and the 'TTL' is set to 'Auto'. At the bottom of the form are 'Cancel' and 'Save' buttons.

Name	Type	Content	Proxy status	TTL	Tags	Comment	Detail
hirewebsites.com	A	201134.99	Proxied	Auto	—	—	Edit

Add record

[name] points to [IPv4 address] and has its traffic proxied through Cloudflare.

Type: A

Name: Use @ for root

IPv4 address:

Proxy status: Proxied

TTL: Auto

Record Attributes

Cancel Save

- b.
4. Wait a few minutes for the DNS to be set

Setting up a certificate / https

1. Go into your VM and do
 - a. Sudo apt install certbot python3-certbot-nginx -y
 - b. Sudo cerbot -nginx -d mydomain.com -d www.mydomain.com
 - c. Enter your email and agree to terms
 - d. Sudo cerbto certificates (ensures its installed and valid)
 - e. Sudo systemctl status certbot.timer (auto renews cert)
 - f. In depth explanation found at <https://certbot.eff.org/>

Using WinSCP

- Since the website needs photos the easiest way to transfer the required files to the VM is through WinSCP
- Download here <https://winscp.net/>
- Setup and install
- Enter details on the first page and then input your private SSH key
 - By clicking on advanced → SSH → Auth → Select private key file
 - **Make sure file protocol is SFTP**

Website script

```
script>
// HERO CAROUSEL
//comments added for documentation explanation

//get carousel track container
const track = document.getElementById('heroTrack');
//convert slide leemtns into an array
const slides = Array.from(track.children);

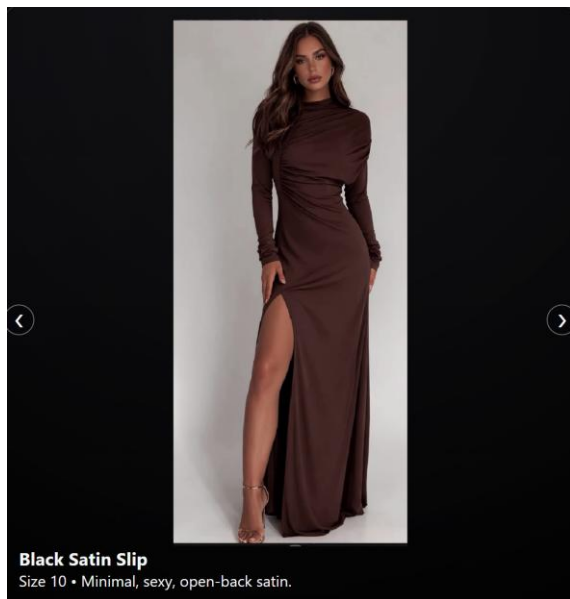
//get navigation buttons
const nextBtn = document.getElementById('nextHero');
const prevBtn = document.getElementById('prevHero');

//display dots under carousel
const dots = Array.from(document.getElementById('heroDots').children);

//track slide index
let index = 0;

//update carousel position and active dot
function updateCarousel() {
  track.style.transform = `translateX(${index * 100}%)`;
  dots.forEach((d, i) => d.classList.toggle('active', i === index));
}

nextBtn.addEventListener('click', () => {
  index = (index + 1) % slides.length;
  updateCarousel();
});
```



Controls the rotating images

Works by

- Finding the div that contains all the images
- Puts the images into an array so the script can cycle through
- Const prev/next btn find the button to be clicked
- Index tracks which slide is currently displayed
- Function that updates the carousel highlighting the correct dot and moving depending on the number
- Interval is set for every 5 seconds so that the images can rotate

Why the script is useful

- Due to its modularity it can take in new photos by just adding photos to the array
- It makes the website appear more professional and dynamic
- Able to display standout clothing options easily
- Customer does not have to scroll down to see the websites offerings